ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. IT ENGINEERING — SEMESTER 6

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain NIST Cloud Computing reference architecture, SPI service models (SaaS, PaaS, IaaS). [14 Marks]

[OR]

Q.1 (Alt) Describe Cloud Deployment Models: Public, Private, Hybrid, and Community clouds with trade-offs. [14 Marks]

--- UNIT II ---

Q.2 Explain Virtualization types: Full virtualization, Para-virtualization, and OS-level containerization (Docker). [14 Marks]

[OR]

Q.2 (Alt) Compare Hypervisors Type-1 (Bare Metal) vs Type-2 (Hosted) with performance benchmarks. [14 Marks]

--- UNIT III ---

Q.3 Explain Distributed Storage: Google File System (GFS) and Hadoop HDFS architecture. [14 Marks]

[OR]

Q.3 (Alt) Describe MapReduce programming model with WordCount example and shuffle/sort phase. [14 Marks]

--- UNIT IV ---

Q.4 Explain Cloud Resource Management: Elasticity, Auto-scaling, and Dynamic Load Balancing algorithms. [14 Marks]

[OR]

Q.4 (Alt) Explain Service Level Agreements (SLAs) and Quality of Service (QoS) metrics in cloud services. [14 Marks]

--- UNIT V ---

Q.5 Describe Cloud Security challenges: Identity Access Management (IAM), Multi-tenancy, and Data Encryption. [14 Marks]

[OR]

Q.5 (Alt) Explain Serverless Computing architecture (AWS Lambda, Google Cloud Functions) and Edge Computing. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****